LXCE on ThinkSystem V3 servers

New features and enhancements

Tool overview

LXCE has the following new features and enhancements for the ThinkSystem V3 platform:

- Version update to support ThinkSystem V3
  - OneCLI: 4.0.1
  - UpdateXpress: 4.0.1
  - BoMC: 13.0.1

- New OS support – Windows 10 and 11
  - OneCLI Binary
    - lnvgy_utl_lxce_oneclixxx-4.0.1_winsrv_x86-64.zip
    - lnvgy_utl_lxceb_oneclixxx-4.0.1_winsrv_x86-64.exe
  - UpdateXpress Binary
    - lnvgy_utl_lxce_uxxxx-4.0.1_windows_x86-64.zip

Always check the Lenovo XClarity Essentials OneCLI (OneCLI) website for the latest OneCLI information and User Guide.

- https://datacentersupport.lenovo.com/us/en/documents/lnvo-tcli

Major OneCLI software feature updates

OneCLI has the following feature updates for the ThinkSystem V3 platform:

- OneCLI bundle update
- OneCLI remote update
- OneCLI out-of-band (OOB) RAID on VROC and Microchip RAID controllers
- OneCLI one time boot configuration
- Miscellaneous new OneCLI functions

OneCLI – Bundle update

The new bundle update option comes with a new `--bundle` parameter and has the following features:

- Update firmware with a bundled package or packages
- OneCLI will send a .zip bundle pack with one or more firmware components to XCC to initiate the firmware update process

A new optional `–applytime` parameter has the following three values:

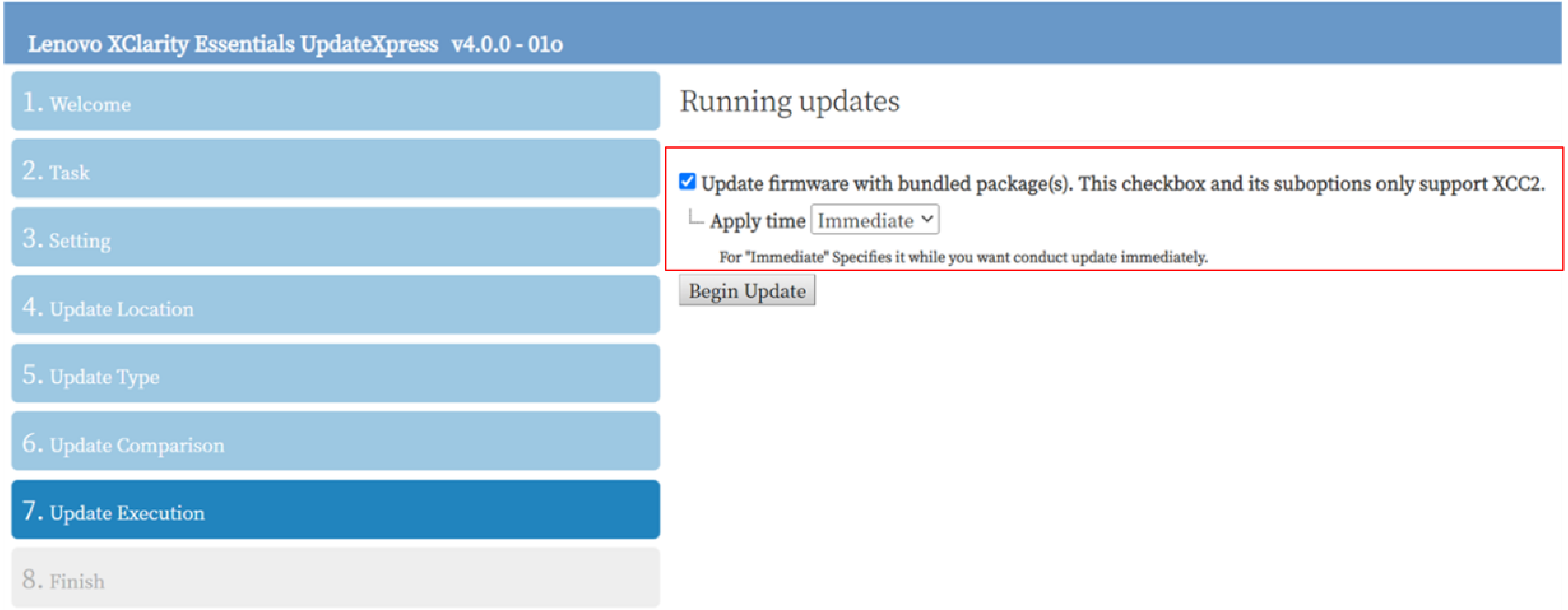
- `OnReset` - default for in-band (IB) mode
- `Immediate` - default for OOB mode
- `OnStartUpdateRequest` - users start the update process manually

The new manual staged update option works with the `checktask` and `startstaged` commands .

- `Checktask` - check staged bundle task status
- `Startstaged` - start staged update task

Bundle update in UpdateXpress

There is a new **Update firmware with bundles package(s)** option in the **Update Execution** section.



Lenovo XClarity Update Manager bundle package

The Lenovo XClarity Update Manager (LXUM) bundle package is a new core firmware component which allows XCC to boot a Bare Metal Update (BMU) environment to carry out iBMU updates for non-agentless I/O firmware like HDD and Persistent Memory (PMems).

For example: *lnvgy_fw_lxum_eal501c-1.00_anyos_comp*

Manual staged update example

In this example, the `--bundle` and `– applytime` parameters are used with the `OnStartUpdateRequest to perform` value.

```
OneCli.exe update flash --bmc "USERID:PASSWORD@10.241.136.145" --bundle  --dir D:\onecli\egs\7d75-shanghai-bundle --applytime OnStartUpdateRequest --includeid lnvgy_fw_drvwn_eal301o-4.00_anyos_comp
```

In the screen capture shown, you can see `Update task staged to BMC successfully` at the prompt, and the task ID is also given:

```
Update task staged to BMC successfully, task id is: 82c05d24-da56-4e60-a112-7be807dca383
```

Flash Results:

No.	Updatable Unit	Slot	Need Reboot	Result	Update ID
1	LXPM Windows Drivers	N/A	Flashing requires additional command	Staged	lnvgy_fw_drvwn_eal301o-4.00_anyos_comp

Then use the `startstaged` command to execute the update:

```
OneCli.exe update startstaged --bmc "USERID:PASSWORD@10.241.136.145"
```

Manual staged update – `checktask`

You can also use `checktask` command to check staged bundle task status before executing the `startstaged` command:

```
OneCli.exe update checktask -b "USERID:PASSWORD@10.241.136.145" --taskid 82c05d24-da56-4e60-a112-7be807dca383
```

```
Task state is: completed
Task associated update job is: JobR000043-Update
Query job JobR000043-Update state...
Query update unit information...

                                   Job State:
   ===================================================================================
   |        ID         | Percent |      StartTime      |       EndTime       |   State   |
   -----------------------------------------------------------------------------------
   | JobR000043-Update | 100     | 2022-09-20 23:36:47 | 2022-09-20 23:37:50 | Completed |
   ===================================================================================

                                                  Job Steps of JobR000043-Update:
   ==================================================================================================================================================
   | No. |      Update Unit      | Percent | Slot |      StartTime      |       EndTime       |   State   |                  Update ID                  |
   --------------------------------------------------------------------------------------------------------------------------------------------------
   | 0   | LXPM Windows Drivers  | 100     | N/A  | 2022-09-20 23:36:47 | 2022-09-20 23:37:46 | Completed | lnvgy_fw_drvwn_eal301o-4.00_anyos_comp.uxz |
   ==================================================================================================================================================
Save to D:\OneCli\bins\lnvgy_utl_lxce_onecli02c-4.0.0_winsrv_x86-64\update\OneCli-Update-Job-status.xml
Succeed.
```

Remote updates

OneCLI has a new feature to support remote updates from Windows to Windows platforms and from Windows to Linux platforms:

- From Windows to Windows
  - `OneCli.exe update flash --remoteos lxce_user:LXCE2021_OneCli@WIN-O8VH5AM4P9U --dir D:\OneCli\bins\fw`
  - As shown in the screen capture, some steps should be completed first
- From Windows to Linux
  - `OneCli.exe update  flash --remoteos root:SYS2022health@10.240.199.19 --dir D:\dd_fw`
  - Linux binary in the specified directory
  - Root privileges are required

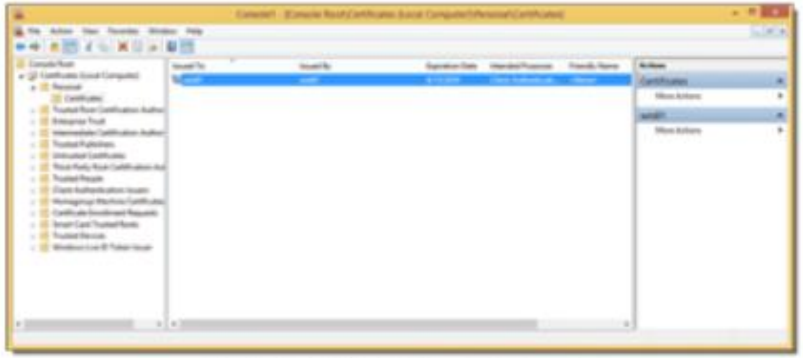
PowerShell 5 Remoting over HTTPS with a self-signed SSL certificate

Enabling HTTPS for PowerShell Remoting

On the remote computer

1. Create a self-signed ssl certificate with hostname or FQDN

```
$Cert = New-SelfSignedCertificate -CertstoreLocation Cert:\LocalMachine\My -DnsName <HostName | DomainName>
```



Certificate in MMC on the remote computer

2. Export the certificate to a file

```
Export-Certificate -Cert $Cert -FilePath C:\temp\cert
```

3. Enable PowerShell Remoting

```
Enable-PSRemoting -SkipNetworkProfileCheck -Force
```

Enable-PSRemoting also starts a WS-Management listener, but only for HTTP. If you want to, you can [illegible] this by reading the contents of the WSMan drive:

```
dir wsman:\localhost\listener
```

OOB RAID on VROC and Microchip in OneCLI

- OOB RAID support on VROC – only supported on V3 servers with Intel processors and NVMe SSDs:
  - `OneCli.exe raid add --file RAID_SW_new.ini --ctrl 3 -b USERID:PASSWORD@10.240.195.209`
  - File: RAID_SW_new.ini

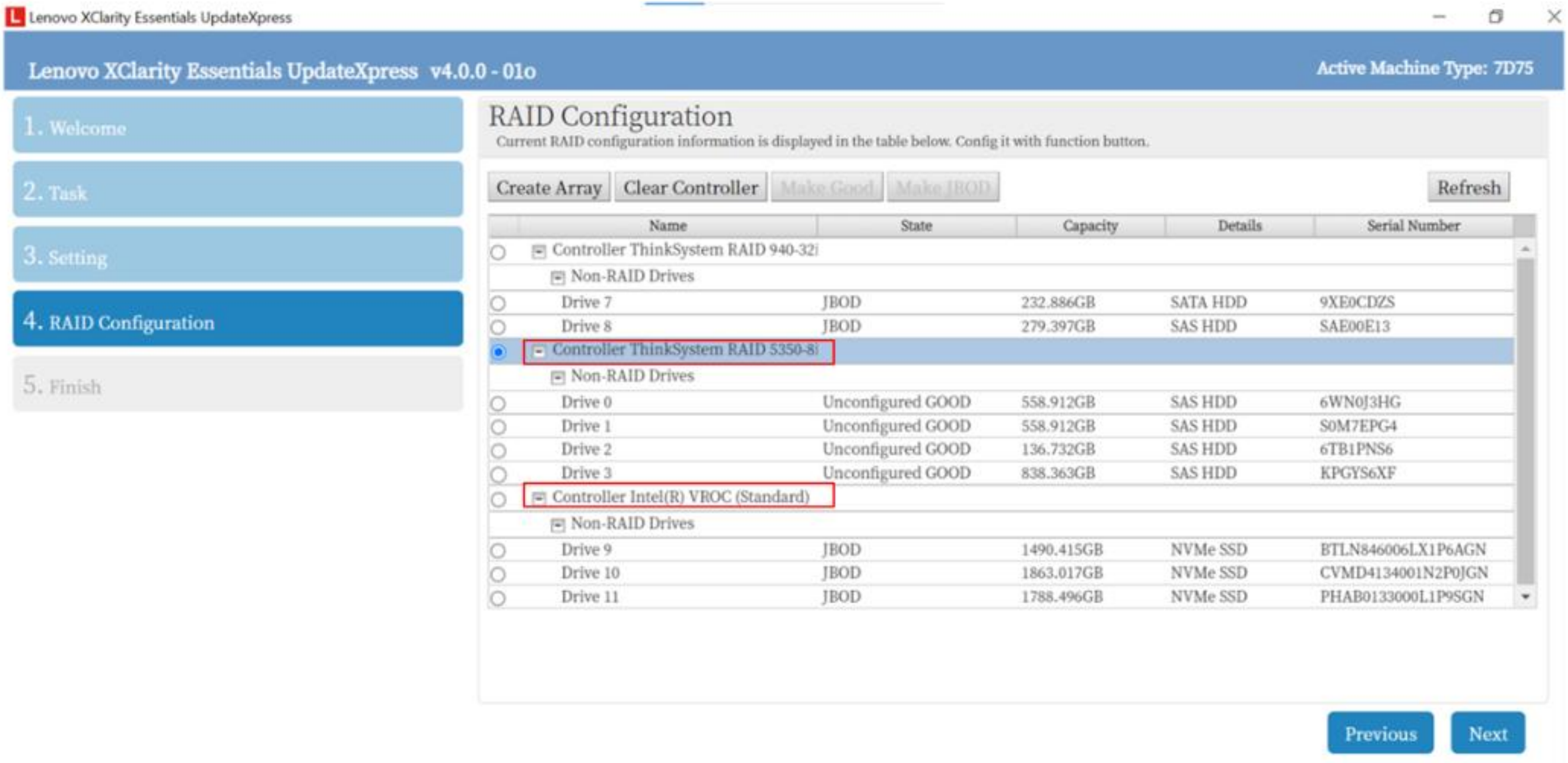
```
[ctrl3-vol0]
  disks=10,11
  raid_level=1
  vol_name=volume1
  volume_size=600GB
  strip_size=128K
```

- OOB RAID support on Microchip – this feature was first available with OneCLI 3.5 and is now also supported on the ThinkSystem V3 platform:
  - `OneCli.exe raid add --file RAID_HW_new.ini  --ctrl 2 -b USERID:PASSWORD@10.240.195.209`
  - File: RAID_HW_new.ini

```
[ctrl2-vol1]
  disks=0,1
  raid_level=1
  vol_name=volume1
  read_policy=0
  write_policy=3
  volume_size=300GB
  strip_size=128K
```

OOB RAID on VROC and Microchip in UpdateXpress

In the RAID Configuration section of UpdateXpress, you can see that Microchip RAID adapters (5350-8i is shown as an example in the screen capture) and Intel VROC are supported.



OOB RAID configuration in UpdateXpress

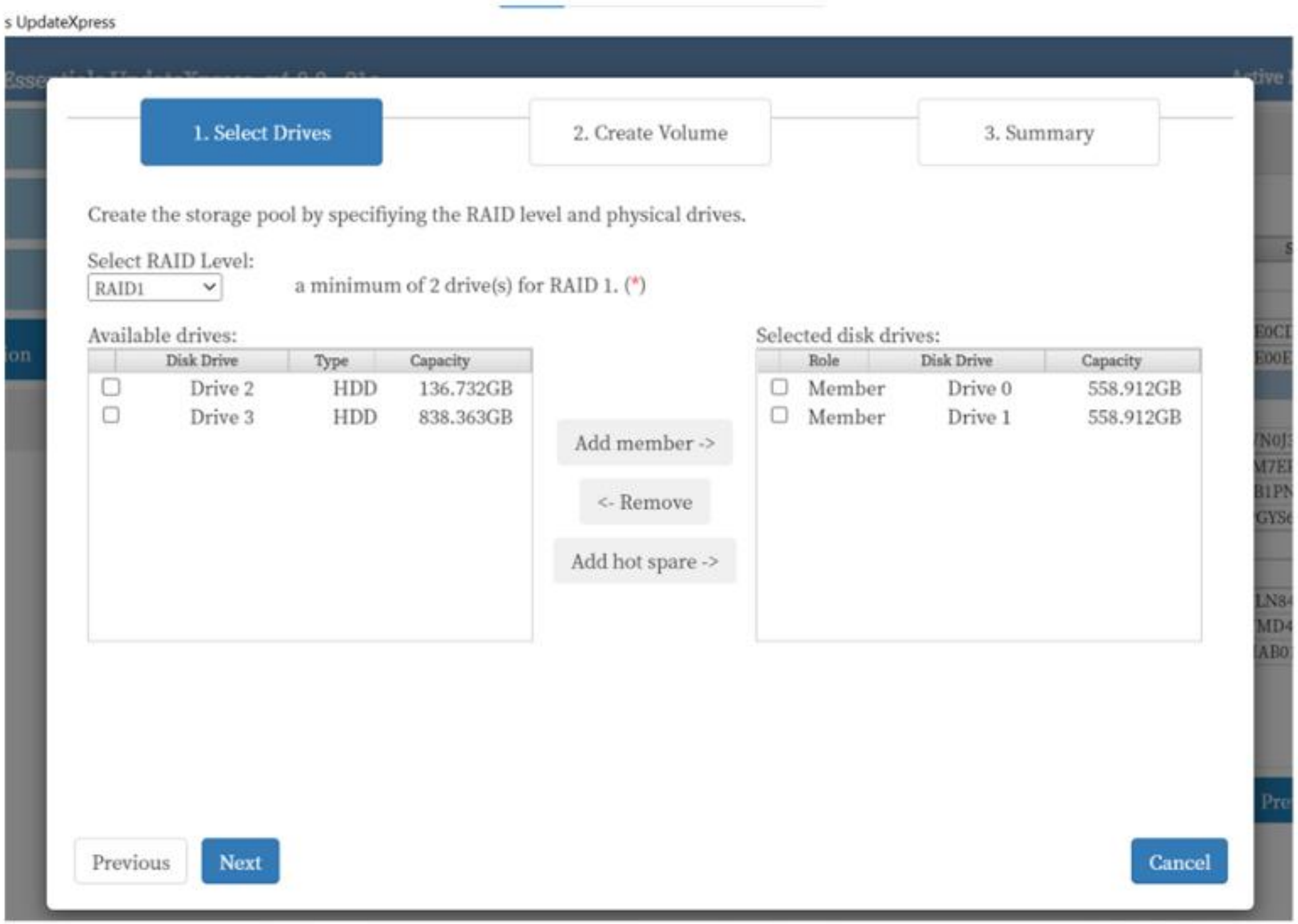
Click each step in turn to see an example of OOB RAID configuration in UpdateXpress

Click each step in turn to see more information

Step 1 2 3

OOB RAID configuration in UpdateXpress

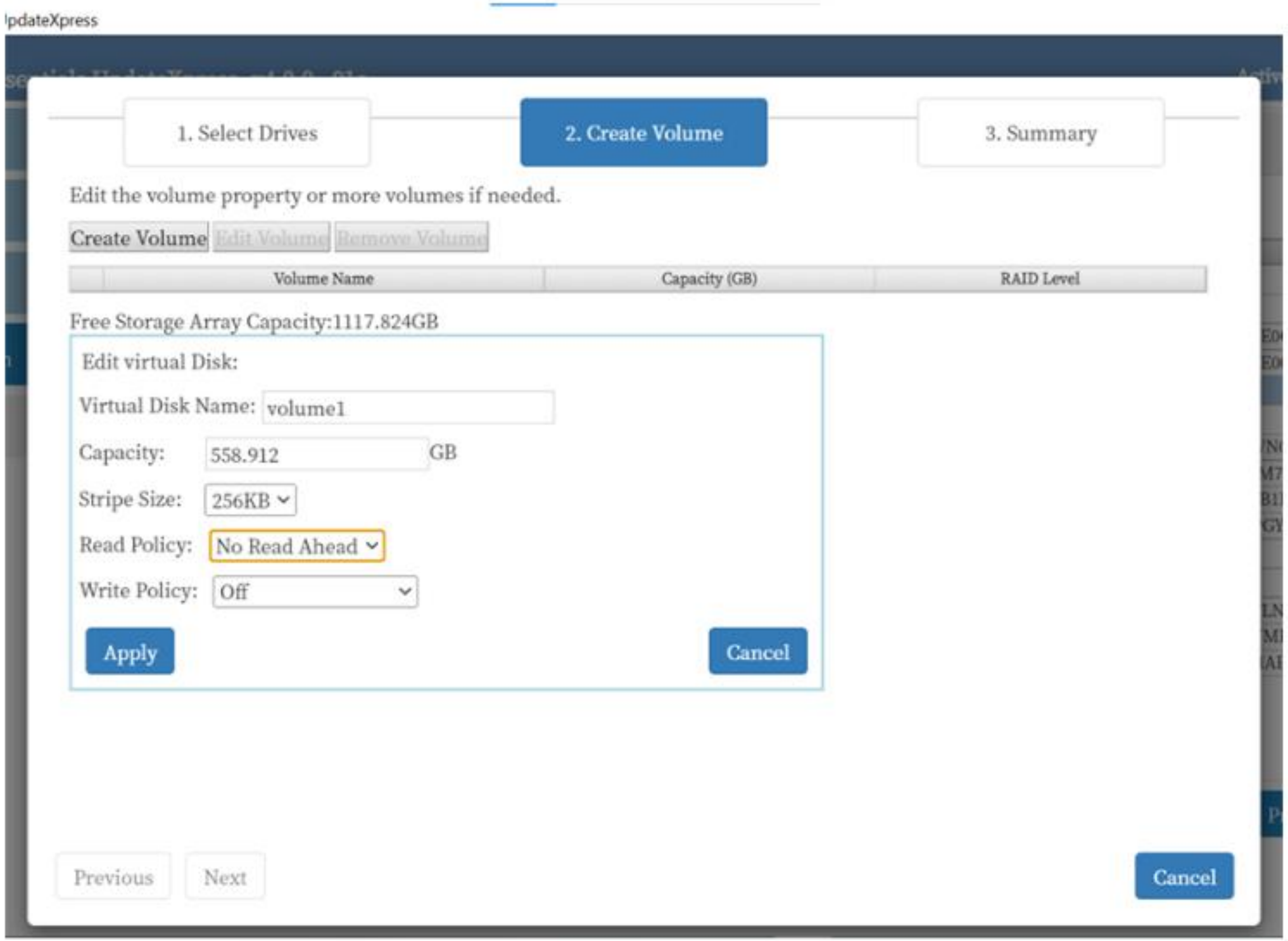
Select drives



Step 1 2 3

OOB RAID configuration in UpdateXpress

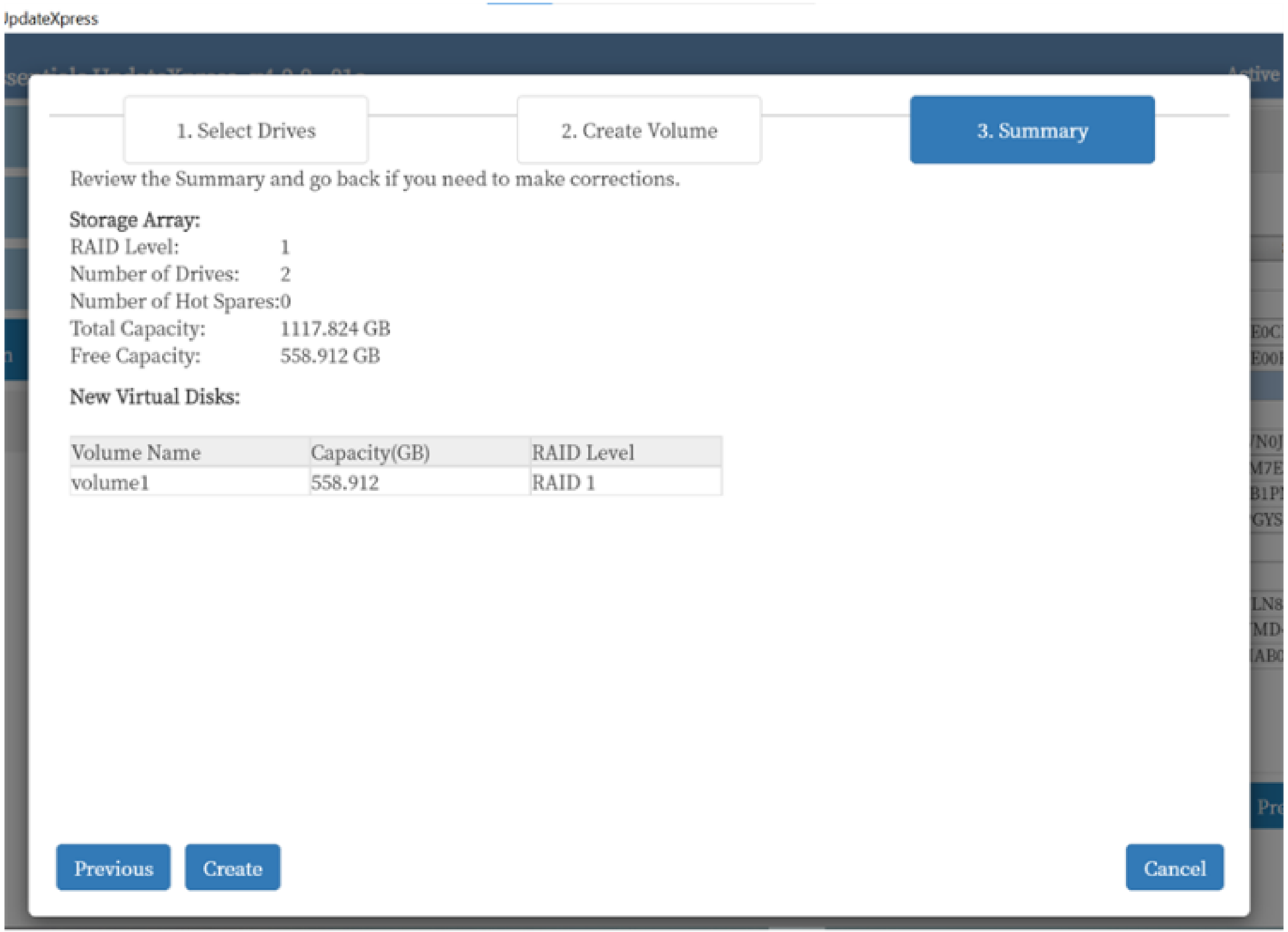
Create a volume



Step 1—2—3

OOB RAID configuration in UpdateXpress

Summary



Step 1 2 3

One time boot configuration

The new `show boot` command is used to see the current system boot mode and boot order:

```
OneCli config show Boot --redfish
```

Possible output examples:

```
Boot.BootSourceOverrideEnabled=Disabled
Boot.BootSourceOverrideMode=UEFI
Boot.BootSourceOverrideTarget=None
Boot.UefiTargetBootSourceOverride=null
```

The `OneTimeBoot` command is used to change the boot mode or boot sequence only for the next boot. In this example, the boot source is changed to the USB drive.

```
OneCli config batch --file OneTimeBoot.txt --redfish -b
USERID:PASSW0RD@10.240.197.46
```

File OneTimeBoot.txt

```
set Boot.BootSourceOverrideEnabled Once
set Boot.BootSourceOverrideMode UEFI
set Boot.BootSourceOverrideTarget Usb
```

Collecting the Service Data Log with OneCLI

The new OneCLI `inventory` command can be used to collect the Service Data log (mini-log). The system must have XCC2 and OneCLI v4.0 or later.

```
OneCli.exe inventory getinfor --servicelog [--bmc userid:pass@bmchost]
```

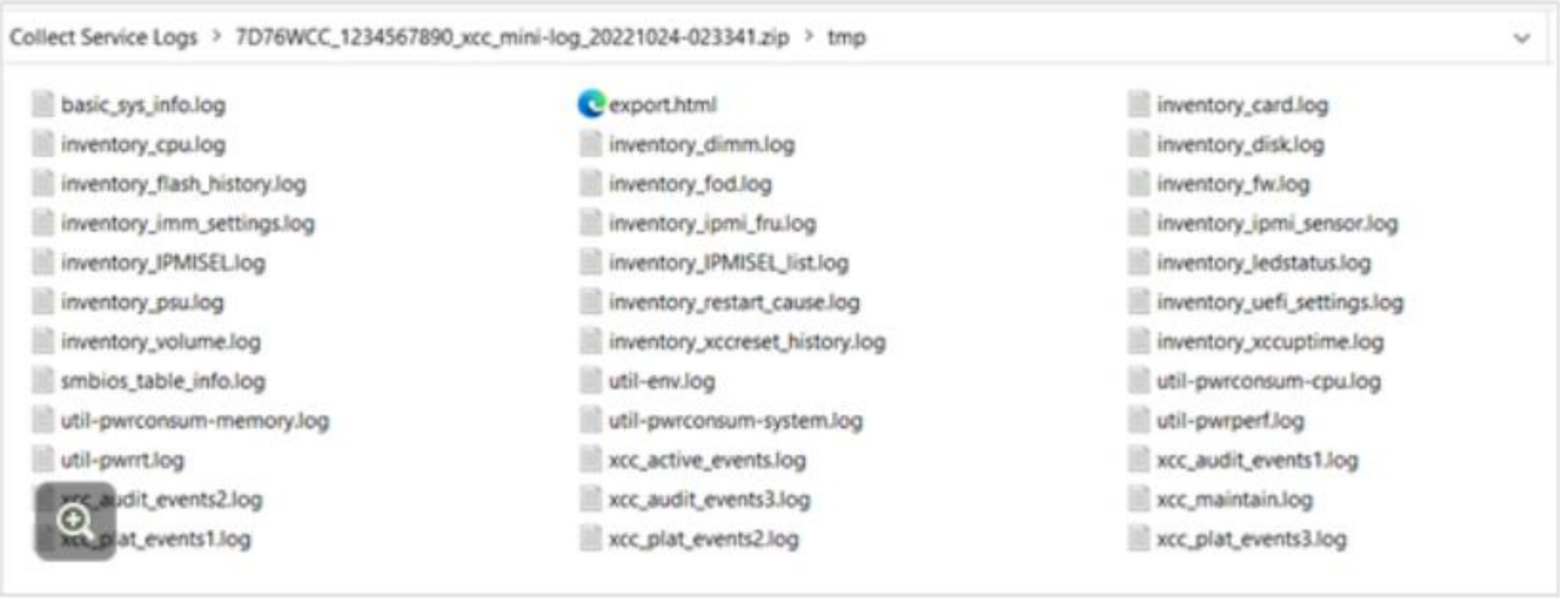
The OneCLI `misc` command can also be used to collect the mini-log

```
OneCli.exe misc servicelog  [--bmc userid:pass@bmchost]
```

The naming convention for the Service Data Log file is:

MachineType+Model_SerialNumber_xcc_mini-log_date_time.zip – for example:

7D76WCC_1234567890_xcc_mini-log_20221024-023341.zip



Collecting service data with OneCLI

The OneCLI `misc servicedata` command, available from OneCLI 3.5 onwards, can be used to get service data information from the BMC.

- `Servicedata`
  - `--type/-T` to specify the type of service data information to get from the BMC – information types are as follows:
    - `osfailure`: to get the latest OS failure screen
    - `healthreport`: to get the health report
    - `all`: to get both the latest OS failure screen and the health report

Sample syntax:

```
OneCli.exe misc servicedata –type all -b USERID:PASSWORD@xxx.xxx.xxx.xxx
```

Example output destination:

\OneCli\bins\lnvgy_utl_lxce_onecli02c-4.0.0_winsrv_x86-64\servicedata\ServiceData-20220921-153132

The latest OS failure screen is a picture saved as an XML file and zipped in a TGZ file – for example: *HealthReport-20220921-153132-56796.xml*

Miscellaneous new functions

`Sysguard` – for System Guard Service management

- `clearsnapshot` – to clear all snapshot data in BMC System Guard

Example syntax:

- `./OneCli sysguard clearsnapshot -b USERID:PASSWORD@10.240.195.209`

`Hostinterface` – Management Host Interface account

- `Clear` – to clear the Host Interface account

Example syntax:

- `./OneCli hostinterface clear --bmc-user USERID –bmc-password PASSWORD`

Miscellaneous new functions - continued

`rotkit` – to generate a boot kit for an RoT upgrade

- `--dir` – specifies the directory for system update packages (SUPs) - if no directory is specified, the current directory will be used
- `--rotdir` – the RoT kit output directory – by default, ./rot_kit/ is used

```
D:\OneCli\bins\lnvgy_utl_lxce_onecli01w-4.0.0_winsrv_x86-64>OneCli.exe rotkit --dir D:\OneCli\EGS\UEFI --rotdir rot_kit

Lenovo XClarity Essentials OneCLI lxce_onecli01w-4.0.0
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OneCLI License Agreement and OneCLI Legal Information can be found at the following location:
    "D:\OneCli\bins\lnvgy_utl_lxce_onecli01w-4.0.0_winsrv_x86-64\Lic"

Querying updates done.
Query results saved to: D:\OneCli\bins\lnvgy_utl_lxce_onecli01w-4.0.0_winsrv_x86-64\rotkit\Onecli-update-query.xml

                                        Selected Packages:
    ==========================================================================================
    | No. | New Version   |         Machine Type           |              Update ID              |
    ------------------------------------------------------------------------------------------
    | 1   | ESE107j-0.70  | 7D72,7D73,7D74,7D75,7D76,7D77  | lnvgy_fw_uefi_ese107j-0.70_anyos_comp |
    ==========================================================================================
Succeed.
```

Directory of rot_kit includes:

```
│   RoTFirmwareManifest.json
│
└─fw
        lnvgy_fw_uefi_ese107j-0.70_anyos_comp.uxz
```

More multi-function commands

Command	Description
`multiflash`	Update firmware requiring an upgrade on multiple remote BMC systems
`multicompare`	Compare firmware requiring an upgrade on multiple remote BMC systems
`multiscan`	Scan firmware requiring an upgrade on multiple remote BMC systems
`multiconfig`	Configure commands for multiple servers
`multiinventory`	Acquire system information from multiple servers
`multisyshealth`	Remotely collect system health from multiple systems
`multiraid`	Remotely configure RAID for multiple systems
`multibmcpassword`	Remotely change the BMC account password for multiple systems
`multiospower`	Operate the OS status – for example, turn on or off – for multiple servers
`multiffdc`	Get BMC/SMM FFDC from multiple servers
`multiservicedata`	Get service data information from multiple BMC accounts
`multivm`	Manage virtual medias on multiple BMC accounts